



Letter to the Editor

Delayed death caused by haematoma after manual strangulation: a rare case

Sir,

We report the case of a 58-year-old man who had an altercation with his neighbour after drinking. He was seized by the neck, and although he did not complain of any discomfort at the time, he was found dead two hours later.

The post mortem examination showed two lacerated bruises on his lips and a longitudinal fracture of the thyroid cartilage plate. There was a 3.0 x 2.0 x 3.0 cm haematoma between the rear part of the thyroid cartilage plate and the annular cartilage space (Fig. 1), which occupied the pyriform recess and obstructed the inlet of the larynx. There were signs of haemorrhage on both sides of the upper and the lower vocal folds, as well as on the retro-oesophageal wall. On histological examination, the laryngopharyngeal mucosa and muscular layers were found to have haemorrhagic

infiltrations associated with a small number of neutrophils and monocytes, particularly the fissure of the glottis (Fig. 2). The right cricothyroid muscle cells showed severe eosinophilic changes and were possibly necrotic. Other histological examinations showed nothing relevant.

Manual strangulation causes asphyxia because of pressure on the airway, the blood vessels, and the nerves in the neck,^{1,2} and in most cases, people die soon after. When they survive, further clinical examinations such as laryngoscopy, magnetic resonance imaging and computed tomography are done only if there is a suspicion of injury to the larynx.³ In this case, the man did not complain of discomfort after the assault, so his



Fig. 1. Haematoma between the rear of the thyroid cartilage plate and the annular cartilage space (black arrow).

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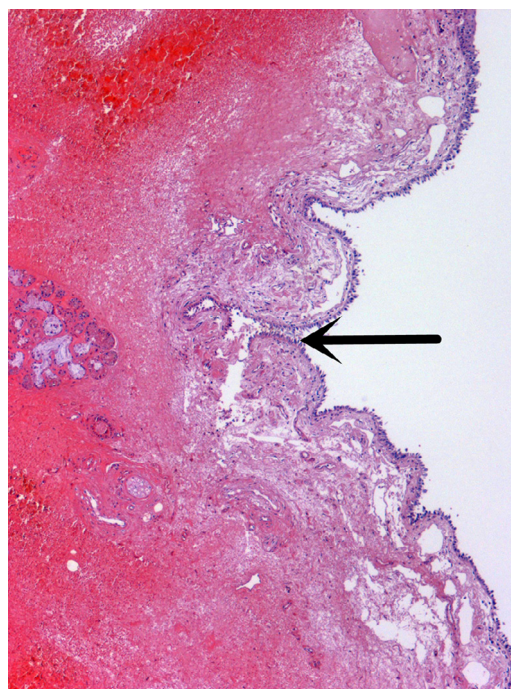


Fig. 2. Haemorrhagic infiltrations associated with a small number of neutrophils and monocytes in the laryngopharyngeal layers of the mucosa and muscles.

neighbor underestimated the importance of his injuries and decided not to take him to the hospital. As the haematoma was large, the blocked laryngeal inlet rapidly deteriorated and his breathing became compromised.

We conclude that as people can survive such an assault for a short time and the internal neck injuries can delay death, it is important for physicians to provide the appropriate care even if there are few signs and symptoms.

Conflict of interest

We have no conflicts of interest.

References

1. Di Paolo M, Guidi B, Bruschini L, et al. Unexpected delayed death after manual strangulation: need for careful examination in the emergency room. *Monaldi Arch Chest Dis* 2009;**71**:132–4.
2. Plattner T, Bolliger S, Zollinger U. Forensic assessment of survived strangulation. *Forensic Sci Int* 2005;**153**:202–7.
3. Yen K, Vock P, Christe A, et al. Clinical forensic radiology in strangulation victims: forensic expertise based on magnetic resonance imaging (MRI) findings. *Int J Legal Med* 2007;**121**:115–23.

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